

ATUL SHARMA

Embedded Systems & IoT Firmware Engineer

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SUMMARY

Embedded Systems and IoT Firmware Engineer with 2+ years of experience writing production Embedded C/C++ firmware for smart meters, GPS cattle trackers, and fintech payment terminals. Work includes Qualcomm cellular IoT (NB-IoT/GPRS), Realtek BLE/Wi-Fi, and STM32/ESP32, plus bare-metal and FreeRTOS drivers, UART/SPI/I2C peripherals, and MQTTS/HTTPS cloud links. Experience in field debugging, OTA/FOTA updates, and hardware-firmware integration for production release.

TECHNICAL SKILLS

Programming Languages: C, C++, Embedded C, Python, Bare-Metal Programming

Protocols & Connectivity: TCP/IP, UDP, HTTP/HTTPS, MQTT/MQTTS, BLE (GAP/GATT), NB-IoT, GPRS, Wi-Fi (802.11 b/g/n), GPS, JSON

Peripherals & Interfaces: UART, SPI, I2C, I2S, GPIO, PWM, ADC, Timers, Watchdog (WDT), USB (Bulk/High-Speed), LCD, Keypad (KPC), Microphone (I2S)

Tools & Build: GCC, ARM GCC, CMake, Makefiles, Git, Keil, IAR, Eclipse, OpenOCD, STM32CubeIDE, STM32CubeMonitor, SCom, Docklight, pyserial

RTOS & Systems: FreeRTOS, CMSIS-OS2, Embedded RTOS, Embedded Linux, POSIX Threads, Ubuntu/WSL2

Microcontrollers & Platforms: STM32 (Cortex-M0/M4), AVR/Arduino, ESP32/8266, Raspberry Pi, Qualcomm Open CPU (NB-IoT/cellular IoT SDK), Qualcomm payment/audio SoC, Realtek (BLE/Wi-Fi), Nordic, Simcom

Security & Data: AES, SHA-256, TLS/SSL, FOTA, NTP, structured logging, onboard file system

WORK EXPERIENCE

Software Engineer — Embedded IoT | MoMAGIC Technologies Pvt. Ltd.

Jul 2024 – Present | New Delhi, India (On-site)

- Architected production firmware in Embedded C/C++ for 3 IoT product lines (smart electricity meters, cattle GPS trackers, fintech audio payment terminals) on Qualcomm cellular Open CPU, Realtek BLE/Wi-Fi, STM32, and ESP32 platforms.
- Deployed Qualcomm NB-IoT/GPRS stacks with TCP/IP sockets, MQTTS/HTTPS publish-subscribe, GPS telemetry, NTP sync, and backoff/reconnect logic for field cellular IoT terminals.
- Developed peripheral drivers and middleware on Qualcomm Open CPU: UART, SPI, I2C, I2S audio codec, GPIO, PWM, ADC, SPI LCD/touch UI, 4x4 keypad, external flash, and low-power sleep modules.
- Delivered fintech payment firmware on Qualcomm payment and audio SoC: 4x4 keypad, I2S microphone, SPI LCD, and on-device QR for UPI/payment UI. Integrated I2S audio for payment tones and voice prompts with HTTPS/MQTTS and AES/SHA-256. Enabled USB bulk transfer for provisioning, log export, and factory diagnostics. Coordinated FreeRTOS multithreading across keypad, LCD, QR, microphone/audio, USB I/O, and cloud connectivity.
- Optimized multithreaded FreeRTOS/CMSIS-OS2 firmware with JSON parsing, AES/SHA-256, and onboard file-system logging for cellular IoT and payment products, improving field diagnostics and cloud payload reliability.
- Verified Realtek BLE/Wi-Fi firmware: GAP/GATT client/server and beacon modes, AP/STA/coexistence, HTU21D sensors, and HTTP/MQTT cloud integration (wireless-only platform).
- Produced Python/C++ UART diagnostic and OTA tools (Tkinter, pyserial) for flashing and production testing, replacing dedicated hardware fixtures used in factory lines.
- Led STM32 bare-metal bring-up with libopenm3: linker scripts, vector tables, Makefile builds, and OpenOCD flashing for register-level debugging.

- Drove code reviews, root-cause analysis on field defects, and firmware hardening for manufacturing release gates across active product deployments.

KEY PROJECTS

Cellular IoT Smart Terminal — Qualcomm Open CPU (NB-IoT SDK) | Jan 2025 – Present

- Released firmware for cellular IoT module: NB-IoT/GPRS registration, SIM/SMS management, MQTTS/HTTPS/TCP cloud links, and GPS location services.
- Integrated SPI touch LCD UI, QR display, keypad input, I2C sensors, FOTA updates, and FreeRTOS multithreaded architecture on Qualcomm Open CPU stack.

Networked Telemetry & Location System (Smart Meters / Cattle Tracking) | Jan 2025 – Present

- Implemented TCP/IP, MQTTS, and HTTPS telemetry with backoff/reconnect logic and structured JSON payloads for smart-meter and cattle-tracking cloud analytics.

Fintech Audio Payment Device Firmware — Qualcomm Payment SoC | Sep 2024 – Present

- Qualcomm payment SoC firmware with 4x4 keypad, I2S microphone, SPI LCD, QR-based UPI UI, USB bulk provisioning, secure MQTTS, and FreeRTOS audio/cloud integration.

Realtek Wi-Fi / BLE IoT Stack Validation | Sep 2024 – Jan 2025

- Verified BLE GAP/GATT, Wi-Fi AP/STA modes, environmental sensor drivers, and HTTP/MQTT cloud integration using IAR and Realtek SDK (wireless-only platform).

Desktop Diagnostic & Configuration Tool | Jul 2024 – Sep 2024

- Authored Python/C++ Tkinter GUI with pyserial for UART diagnostics, firmware flashing, and device configuration used by production and field-support teams.

STM32 Bare-Metal & Raspberry Pi 4 Embedded Drivers | Jan 2024 – Jun 2024

- Implemented STM32 bare-metal boot flow with custom linker scripts and Makefile toolchain; Linux GPIO/I2C/SPI drivers for Raspberry Pi 4 validated with logic analyzer and scope.

EDUCATION

Bachelor of Technology — Electronics & Communication Engineering | CGPA: 8.05 (First Division with Distinction)

Feroze Gandhi Institute of Engineering & Technology, Raebareli | Dr. A.P.J. Abdul Kalam Technical University, Lucknow | 2020 – 2024

CERTIFICATIONS

- Introduction to Cybersecurity — Cisco | Aug 2023
- Microsoft Certified: Azure Fundamentals — Microsoft | Jun 2023
- Microsoft Certified: Power Platform Fundamentals — Microsoft | Jun 2023
- Full Stack JavaScript (MERN) — Fynd | Jul 2023
- Codekaze — Coding Ninjas | Jun 2023
- Web Development — Internshala Trainings | Oct 2022